

Refraction and Total Internal Reflection

The absolute refractive index, Snell's law, the critical angle, optical fibres and the blocking disc.

Grade 11 · Egyptian Baccalaureate

30 questions

Step-by-step solutions

 $\sin \theta_c = n_2 / n_1$

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PHYSICS

The toolkit for this lesson

$$n = c / v$$

the absolute refractive index

$$n_1 \sin \theta_1 = n_2 \sin \theta_2$$

Snell's law of refraction

$$\sin \theta_c = n_2 / n_1$$

the critical angle, $n_1 > n_2$

$$\sin \theta_c = 1 / n$$

when the outside medium is air

$$\lambda_1 / \lambda_2 = v_1 / v_2 = n_2 / n_1$$

f never changes

$$r = h \tan \theta_c$$

the radius of the blocking disc

A denser medium is a slower medium. $n = c/v$, so a bigger n always means a smaller v , a shorter λ and a ray that bends **towards** the normal. Going the other way — out of the denser medium — the ray bends away from the normal, and once the angle inside passes θ_c it cannot get out at all.

CLASSWORK 1

A

Refraction between two media

- 1 The opposite diagram shows a light ray passing between two media A and B. When the ray passes from medium A into medium B, **what happens to the speed of light and to the wavelength?**

EXPLAIN

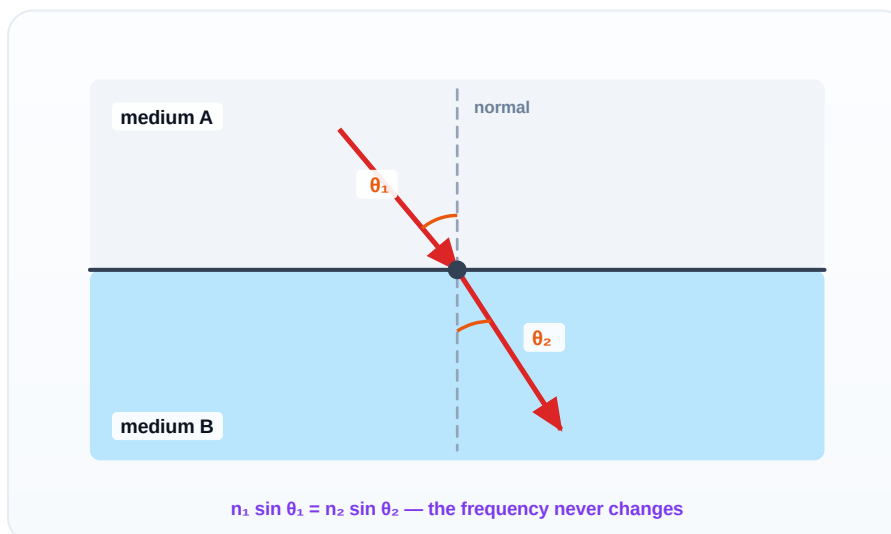


FIGURE 1

2 The absolute refractive index (n) of a transparent medium is given by the relation

EXPLAIN

HOME ASSIGNMENT 1

B The refractive index

3 Light falls from water into glass with an angle of incidence of 55° . Given that the relative refractive index between water and glass equals 1.15, **calculate the angle of refraction of the light in the glass.**

PROBLEM

4 **Compare between** the optically denser medium and the optically less dense medium as regards: the speed of light in it, and the absolute refractive index.

EXPLAIN

5 If the speed of light in air is 3×10^8 m/s and its speed in glass is 2×10^8 m/s, **calculate the absolute refractive index of the glass.**

PROBLEM

6 When light passes from a less optically dense medium into a more optically dense medium, then

MCQ

A both the frequency and the wavelength decrease

B the frequency increases and the wavelength decreases

C the frequency does not change and the wavelength increases

D the frequency does not change and the wavelength decreases

- 7 A light ray falls from medium (1), whose absolute refractive index is 1.3, into medium (2), whose absolute refractive index is 1.5, as in the figure. Which of the choices shows what will happen to both the wavelength and the speed of light in medium (2)?

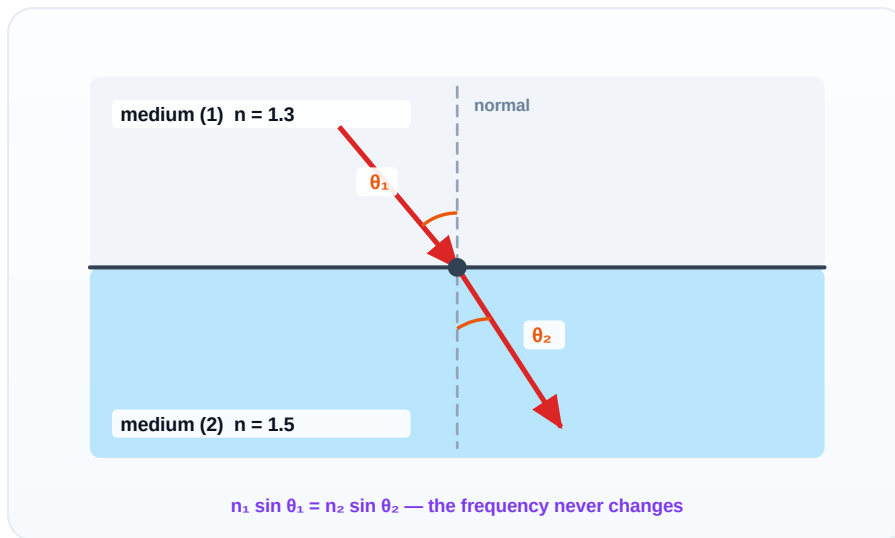


FIGURE 2

- A both the wavelength and the speed increase
- B both the wavelength and the speed decrease
- C the wavelength increases and the speed decreases
- D the wavelength decreases and the speed increases

CLASSWORK 2

C The critical angle

- 8 When light passes from a more optically dense medium into a less dense medium, **what is the greatest value of the angle of refraction** in the less dense medium?

EXPLAIN

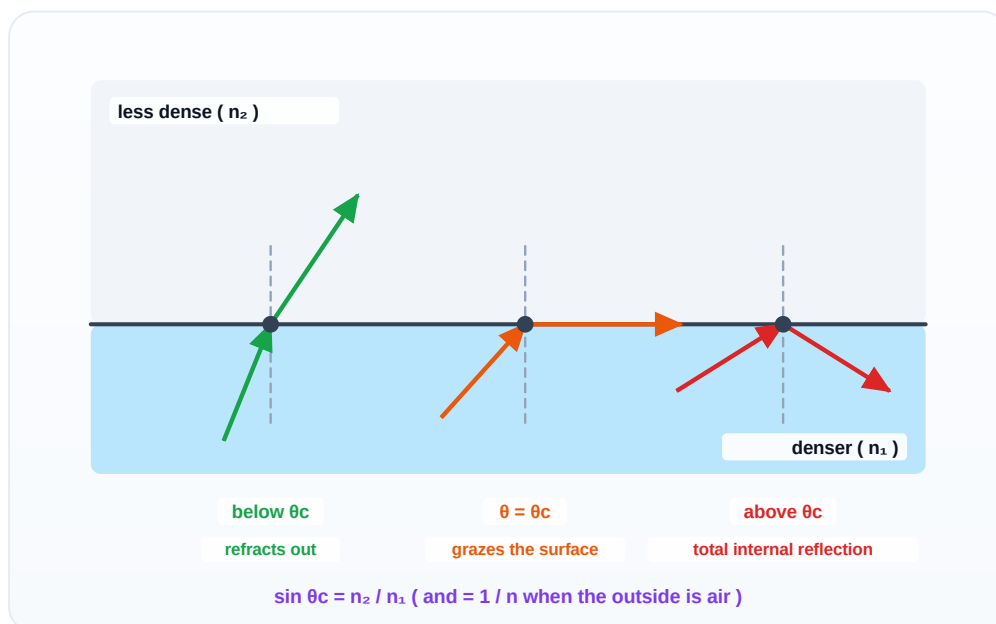


FIGURE 3

9

To calculate the critical angle between two media ($n_1 > n_2$) we use the relation

[EXPLAIN](#)

Total internal reflection

- 10 **Give the scientific reason:** the outer cladding of an optical fibre is made of a material whose refractive index is less than the refractive index of the inner core.

EXPLAIN

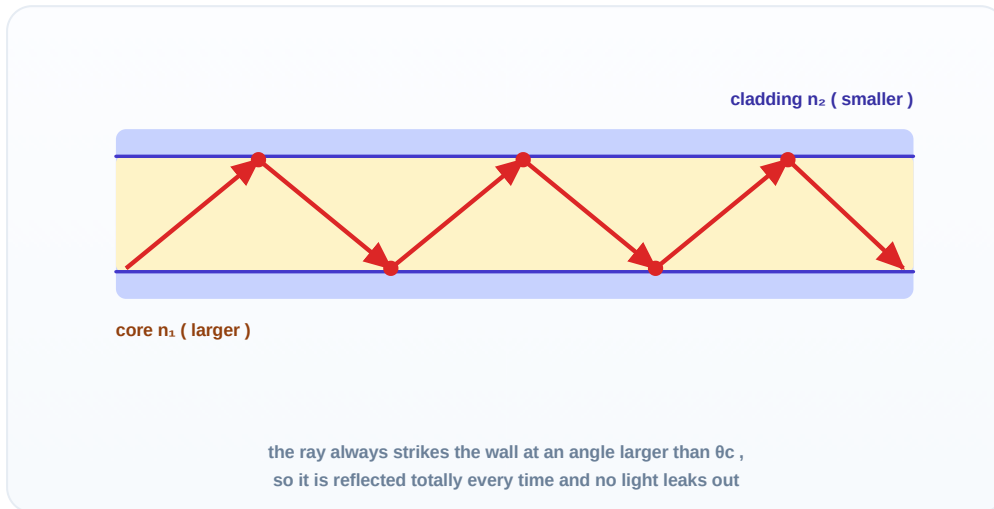


FIGURE 4

- 11 **What happens when** a narrow beam of light is directed upwards through the side of a glass of water towards the (water – air) surface, and the angle of incidence is gradually increased?

EXPLAIN

- 12 If the refractive index of the core in an optical fibre is 1.6 and that of the cladding is 1.4, then $\sin \theta_c$ between them equals:

MCQ

A 1.4 / 1.6

B 1.6 / 1.4

C 1 / 1.6

D 1 / 1.4

- 13 Total internal reflection occurs when light passes from a medium:

MCQ

A of lower optical density into a medium of higher optical density

B of higher optical density into a medium of lower optical density

C into another of equal density

D that is opaque into one that is transparent

- 14 Given that the refractive index of glass is 1.5, which of the following figures shows the correct path of a light ray falling on the boundary separating the glass and the air at an angle of 50° ?

MCQ

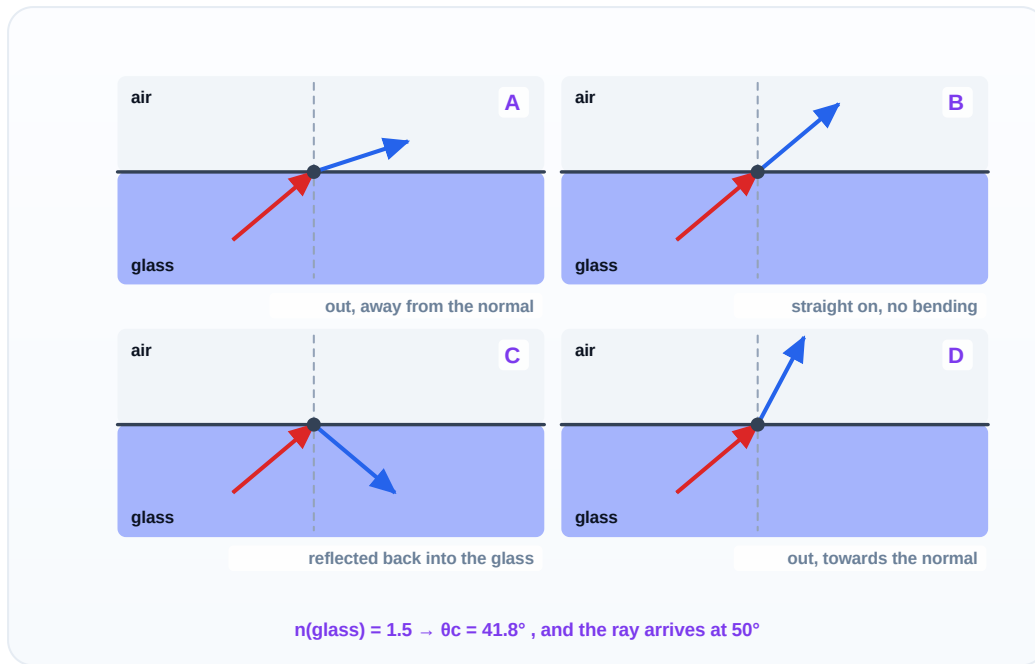


FIGURE 5

A figure A

B figure B

C figure C

D figure D

WEEKLY ASSESSMENT

E

Group A

- 15 An optical fibre whose material has a refractive index of 2.1 is covered with an outer layer of cryolite. The refractive index of the outer layer that makes the critical angle between the two layers equal 32° is

MCQ

A 1.11

B 2.25

C 3.96

D 4.32

- 16 In the experiment of blocking light with a disc, if the water is replaced by another liquid of greater refractive index, then the area of the disc needed for the blocking:

MCQ

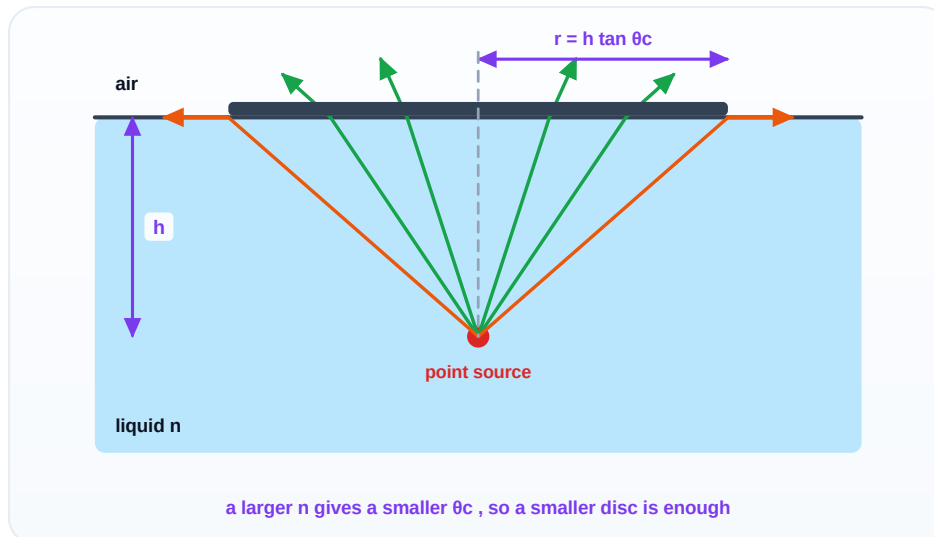


FIGURE 6

- A increases
- B decreases
- C does not change
- D becomes zero

- 17 Give the scientific reason: a light ray bends away from the normal when it passes from water into air.

EXPLAIN

- 18 What happens when a light ray falls from a more optically dense medium into a less optically dense medium at an angle of incidence equal to the critical angle?

EXPLAIN

- 19 Light whose speed in vacuum is c passes into two media A and B; its speed in A is $0.8c$ and in B is $0.6c$. Calculate the critical angle when light passes between the two media.

PROBLEM

WEEKLY ASSESSMENT

F Group B

- 20 A light ray falls on the boundary separating two media whose absolute refractive indices are respectively n_1 and n_2 , where $n_1 > n_2$. What is the correct choice that expresses the condition required for total internal reflection to occur?

MCQ

- A the ray must fall in the medium n_1 at an angle greater than the critical angle
- B the ray must fall in the medium n_2 at an angle greater than the critical angle
- C the ray must fall in the medium n_1 at an angle less than the critical angle
- D the ray must fall in the medium n_2 at an angle less than the critical angle

21 If a light ray falls on a rectangular glass block with an angle of incidence equal to 60° , and the absolute refractive index of the glass = $\sqrt{3}$, then the angle of refraction of the light equals

MCQ

A 90°

B 60°

C 30°

D 45°

22 **What happens when** a light ray falls from a more optically dense medium into a less optically dense medium at an angle of incidence greater than the critical angle?

EXPLAIN

23 **Compare between** the more optically dense medium and the less optically dense medium as regards: the speed of light in it, and the absolute refractive index.

EXPLAIN

24 **Give the scientific reason:** the reflected ray inside the water does not disappear when light passes from water into air; rather it increases in brightness as the angle of incidence increases.

EXPLAIN

25 A light ray passes from a material into air. If the angle of incidence is 30° and the angle of refraction is 50° , **calculate the refractive index of the material.**

PROBLEM

WEEKLY ASSESSMENT

G

Group C

26 If a light ray passes from a less optically dense medium into a medium of higher optical density, then the ratio between the wavelength of the incident light and the wavelength of the refracted light is

MCQ

A less than one

B greater than one

C equal to one

D zero

27 When light passes from a more optically dense medium into a less dense medium, then the greatest value of the angle of refraction in the less dense medium is

MCQ

A 180°

B 90°

C 42°

D 45°

28 **Compare between** the passage of light from air into water and its passage from water into air as regards: the frequency, and the wavelength.

EXPLAIN

29

Give the scientific reason: the radius of the disc needed to block the light of a point source under the surface of a liquid decreases as the refractive index of that liquid increases.

EXPLAIN

30

If the critical angle of glass with air is 42° , and the critical angle of water with air is 49° , **which of them is optically denser? And why, mathematically?**

PROBLEM

Solutions — step by step

Everything follows from $n = c/v$. Decide first which medium is the denser one; then the speed, the wavelength and the direction of bending all follow automatically. And remember that θ_c only exists for light trying to **leave** the denser medium.

Part A — Classwork 1

1 passing from medium A into medium B

the figure shows the ray bending **towards** the normal in B, so B is the optically denser medium :

$$n(B) > n(A) \quad \text{and} \quad n = c / v \quad \rightarrow \quad v(B) < v(A)$$

the frequency is set by the source and cannot change at a boundary, so :

$$\lambda = v / f \quad \rightarrow \quad \lambda(B) < \lambda(A)$$

The three always move together: slower, shorter, bent towards the normal. The one quantity that never moves is the frequency — the boundary cannot create or destroy wave crests.

	optically denser medium	optically less dense medium
the speed of light v	smaller	larger
the absolute index $n = c/v$	larger	smaller
the wavelength λ	shorter	longer
the frequency f	the same	the same
a ray entering it	bends towards the normal	bends away from the normal

both the speed of light and the wavelength decrease in B, while the frequency stays the same

2 the definition of the absolute refractive index

the absolute refractive index of a medium is the ratio of the speed of light in vacuum to its speed in that medium :

$$n = c / v$$

it can also be written from the law of refraction, for light going from vacuum :

$$n = \sin(\text{angle in vacuum}) / \sin(\text{angle in the medium})$$

and, since f is unchanged, $n = \lambda(\text{vacuum}) / \lambda(\text{medium})$

Because nothing travels faster than c , v is always less than or equal to c and so $n \geq 1$. For air $n \approx 1.0003$, which is why air is treated as vacuum in school problems.

$n = c / v$ — a pure number, never less than 1

Part B — Home assignment 1

3 water into glass at 55°

the relative refractive index from water to glass is :

$$n(\text{water} \rightarrow \text{glass}) = \sin i / \sin r = 1.15$$

$$\sin r = \sin 55^\circ / 1.15$$

$$\sin r = 0.8192 / 1.15 = 0.7123$$

$$r = 45.4^\circ \approx 45^\circ$$

The relative index is greater than 1, so the glass is the denser of the two and the ray must bend **towards** the normal — and indeed 45° is smaller than 55° . If your answer had come out bigger than 55° you would know the ratio had been used upside down.

$$r \approx 45^\circ$$

4 denser compared with less dense

$n = c / v \rightarrow$ n and v always move in **opposite** directions

	optically denser medium	optically less dense medium
the speed of light v	smaller	larger
the absolute index $n = c/v$	larger	smaller
the wavelength λ	shorter	longer
the frequency f	the same	the same
a ray entering it	bends towards the normal	bends away from the normal

the denser medium has the smaller speed of light and the larger refractive index

5 the refractive index of the glass

$$n = c / v$$

$$n = (3 \times 10^8) / (2 \times 10^8)$$

$$n = 1.5$$

A refractive index is a ratio of two speeds, so it has no unit. 1.5 is the standard value for ordinary glass, and it makes the critical angle with air about 41.8° — a number worth remembering.

$$n = 1.5$$

6 entering a denser medium

the frequency is fixed by the source : f unchanged

the denser medium is the slower one : v decreases

$$\lambda = v / f \rightarrow \lambda \text{ decreases as well}$$

D) the frequency does not change and the wavelength decreases

7 from $n = 1.3$ into $n = 1.5$

$$n_2 = 1.5 > n_1 = 1.3 \quad \rightarrow \text{medium (2) is the denser one}$$

$$n = c / v \quad \rightarrow \quad v_2 < v_1$$

$$\lambda = v / f \text{ (} f \text{ fixed)} \rightarrow \lambda_2 < \lambda_1$$

Note on the printed question. In the book the four choices are given as small diagrams rather than as sentences; they have been written out here in words, with the same meaning, so that the question can be answered from the text alone.

B) both the wavelength and the speed decrease

Part C — Classwork 2

8 the greatest angle of refraction

going into the **less** dense medium the ray bends away from the normal, so the angle of refraction is always larger than the angle of incidence.

the refracted ray can at most run along the surface itself :

$$\text{greatest angle of refraction} = 90^\circ$$

and the angle of incidence that produces it is the **critical angle** θ_c .

Beyond that there is no refracted ray at all: every ray arriving at more than θ_c is reflected back into the denser medium, which is total internal reflection.

90° — the refracted ray grazes along the surface

9 the relation for the critical angle

at the critical angle the refracted ray grazes the surface, so $\theta_2 = 90^\circ$:

$$n_1 \sin \theta_c = n_2 \sin 90^\circ = n_2$$

$$\sin \theta_c = n_2 / n_1 \quad (n_1 > n_2)$$

and when the second medium is air ($n_2 \approx 1$) :

$$\sin \theta_c = 1 / n$$

$\sin \theta_c = n_2 / n_1$ ($= 1/n$ with air outside)

Part D — Home assignment 2

10 why the cladding has the smaller refractive index

total internal reflection can only happen when light tries to leave the **denser** medium, so the core must be the denser of the two :

$$n(\text{core}) > n(\text{cladding})$$

then a critical angle exists at the core-cladding boundary, and every ray that strikes it at more than θ_c is reflected back into the core.

If the cladding were the denser one there would be **no** critical angle at all, and light would escape at every bounce. The cladding also keeps the core surface clean and prevents light from leaking into a neighbouring fibre.

so that a critical angle exists at the core-cladding boundary and the light is totally reflected instead of leaking out

11 raising the angle inside a glass of water

the light is going from water (denser) to air (less dense) :

small angles → most of the light refracts out, bending away from the normal,
and a weak reflected ray stays in the water

$\theta = \theta_c \approx 49^\circ$ → the refracted ray grazes along the surface (90°) and the
reflected ray is at its brightest so far

$\theta > \theta_c$ → the refracted ray disappears completely : **total internal reflection**, all the light stays in the water

the refracted ray bends further and further away until, at θ_c , it grazes the surface and then vanishes — beyond θ_c all the light is reflected back into the water

12 $\sin \theta_c$ for a 1.6 / 1.4 fibre

$$\sin \theta_c = n_2 / n_1 \quad (n_1 = \text{core} , n_2 = \text{cladding})$$

$$\sin \theta_c = 1.4 / 1.6$$

$$\sin \theta_c = 0.875 \quad \rightarrow \quad \theta_c = 61^\circ$$

The smaller index always goes on top — otherwise the sine would come out greater than 1, which is impossible, and that is exactly how you can spot the wrong choices (B) and (D).

A) 1.4 / 1.6

13 when total internal reflection happens

two conditions are needed together :

- 1 the light must be going from the **denser** medium to the less dense one
- 2 the angle of incidence must be **greater** than the critical angle

B) from a medium of higher optical density into a medium of lower optical density

14 the correct path at 50° in glass

find the critical angle for glass and air :

$$\sin \theta_c = 1 / n = 1 / 1.5 = 0.6667$$

$$\theta_c = 41.8^\circ$$

the ray arrives at 50° , and $50^\circ > 41.8^\circ$:

→ no light gets out — the ray is reflected back into the glass, with the angle of reflection equal to the angle of incidence.

Figures A and D would need a refracted ray to exist, which it does not above θ_c , and B would need the two media to have the same index. **Note on the printed question:** the four choices in the book are diagrams without letters; they are drawn here as four labelled panels so that the answer can be given as a letter.

C) — the ray is totally reflected back into the glass at 50°

15 the cryolite cladding for $\theta_c = 32^\circ$

$$\begin{aligned}\sin \theta_c &= n_2 / n_1 & \rightarrow & n_2 = n_1 \sin \theta_c \\ n_2 &= 2.1 \times \sin 32^\circ \\ n_2 &= 2.1 \times 0.5299 \\ n_2 &= 1.11\end{aligned}$$

The other three choices are all **larger** than 2.1, and a cladding denser than the core would make total internal reflection impossible — so they can be rejected before any arithmetic is done.

A) $n = 1.11$

16 a liquid of greater refractive index

$$\begin{aligned}\sin \theta_c &= 1 / n & \rightarrow & \text{a larger } n \text{ gives a smaller } \theta_c \\ r &= h \tan \theta_c & \rightarrow & \text{a smaller } \theta_c \text{ gives a smaller } r \\ \text{area} &= \pi r^2 & \rightarrow & \text{and therefore a smaller area}\end{aligned}$$

The escaping light is confined to a cone of half-angle θ_c . Making the liquid denser narrows that cone, so a smaller lid is enough to cover it.

B) it decreases

17 why a ray bends away going from water to air

$$\begin{aligned}n(\text{water}) &\approx 1.33 > n(\text{air}) \approx 1 \\ n &= c / v & \rightarrow & v(\text{air}) > v(\text{water})\end{aligned}$$

the light speeds up as it enters the air. from $n_1 \sin \theta_1 = n_2 \sin \theta_2$:

$$\sin \theta_2 = (n_1 / n_2) \sin \theta_1 > \sin \theta_1 \quad \rightarrow \quad \theta_2 > \theta_1$$

A picture that helps: think of the wavefront as a line of marchers. The end that reaches the faster medium first pulls ahead, and the whole line swings round — away from the normal when the new medium is the faster one.

because air is the optically less dense medium, so the light travels faster in it and the ray turns away from the normal

18 incidence exactly at the critical angle

$$n_1 \sin \theta_c = n_2 \sin \theta_2 \quad \text{with } \theta_2 = 90^\circ$$

the refracted ray leaves along the surface itself, at 90° to the normal, and is extremely faint; at the same time the reflected ray inside the denser medium is at its brightest so far.

the refracted ray grazes along the surface at 90° — the borderline case just before total internal reflection

19 the critical angle between A and B

$$n = c / v :$$

$$n(A) = c / 0.8c = 1.25$$

$$n(B) = c / 0.6c = 1.667$$

B is the denser one, so the light must be going from B into A :

$$\sin \theta_c = n(A) / n(B) = 1.25 / 1.667$$

$$\sin \theta_c = 0.75$$

$$\theta_c = 48.6^\circ$$

Notice the short cut: $\sin \theta_c = n(A)/n(B) = v(B)/v(A) = 0.6c/0.8c = 0.75$, so the speeds can be used directly without working out either index.

$$\theta_c \approx 48.6^\circ$$

Part F — Weekly assessment, Group B

20 the condition for total internal reflection

the light must start in the **denser** medium, which is n_1 here, and must arrive at the boundary at more than the critical angle :

$$\text{the ray is in } n_1 \quad \text{and} \quad \theta > \theta_c$$

Choices (B) and (D) put the ray in the **less** dense medium, where no critical angle even exists; choice (C) has the inequality the wrong way round, and would simply give ordinary refraction.

A) the ray must fall in the medium n_1 at an angle greater than the critical angle

21 refraction into glass of $n = \sqrt{3}$ at 60°

$$n = \sin i / \sin r \quad \rightarrow \quad \sin r = \sin i / n$$

$$\sin r = \sin 60^\circ / \sqrt{3} = 0.8660 / 1.7321$$

$$\sin r = 0.5$$

$$r = 30^\circ$$

Choice (A), 90° , is impossible for light **entering** a denser medium, and (B) would mean no bending at all. The ray must bend towards the normal, so the answer has to be less than 60° .

C) 30°

22 incidence greater than the critical angle

above θ_c the law of refraction would need $\sin \theta_2 > 1$, which is impossible, so no refracted ray can exist :

→ all the light is reflected back into the denser medium, obeying the ordinary law of reflection (angle of reflection = angle of incidence).

This is the only kind of reflection that is complete. An ordinary mirror always absorbs a few per cent, which is why optical fibres and reflecting prisms in binoculars use total internal reflection instead of mirrors.

total internal reflection takes place — 100 % of the light is reflected back inside the denser medium

23 denser compared with less dense, again

$$n = c / v \quad \rightarrow \quad \text{larger } n \leftrightarrow \text{smaller } v$$

	optically denser medium	optically less dense medium
the speed of light v	smaller	larger
the absolute index $n = c/v$	larger	smaller
the wavelength λ	shorter	longer
the frequency f	the same	the same
a ray entering it	bends towards the normal	bends away from the normal

the denser medium: smaller v , larger n . the less dense medium: larger v , smaller n

24 why the internally reflected ray gets brighter

at every boundary the light divides into a refracted part and a reflected part. as θ grows towards θ_c , less and less energy goes into the refracted ray, so more and more is left for the reflected one. at $\theta = \theta_c$ the refracted ray vanishes and beyond it **all** the energy is reflected.

So the reflected ray was always there; it is simply too faint to notice at small angles. This is the same effect that makes a lake look transparent when you look straight down and like a mirror when you look across it at a glancing angle.

because the share of the energy taken by the refracted ray falls as the angle grows, so the share left for the reflected ray rises — reaching 100 % at and beyond θ_c

25 the index from 30° inside to 50° in air

the light goes from the material into air :

$$n \sin i = n(\text{air}) \sin r = 1 \times \sin r$$

$$n \times \sin 30^\circ = \sin 50^\circ$$

$$n \times 0.5 = 0.7660$$

$$n = 1.53$$

The ray bent away from the normal ($30^\circ \rightarrow 50^\circ$), so the material must be the denser one and n must come out greater than 1 — a quick check that the fraction was not written upside down.

$$n \approx 1.53$$

Part G — Weekly assessment, Group C

26 the ratio of the two wavelengths

going into a denser medium the light slows down, so the wavelength shortens :

$$\lambda(\text{incident}) / \lambda(\text{refracted}) = v_1 / v_2 = n_2 / n_1$$

and $n_2 > n_1$, so the ratio is **greater than one**.

B) greater than one

27 the greatest angle of refraction

the refracted ray bends away from the normal and can at most lie along the surface itself :

greatest angle of refraction = 90°

Choice (C), 42° , is the critical angle of glass with air — an angle of **incidence**, not of refraction. It is put there to catch anyone answering from memory.

B) 90°

28 air to water compared with water to air

the frequency is fixed by the source in both directions :

f is the same going either way

the wavelength follows the speed :

air $\rightarrow$ water : v decreases $\rightarrow \lambda$ decreases , ray bends towards the normal

water $\rightarrow$ air : v increases $\rightarrow \lambda$ increases , ray bends away from the normal

Note on the printed questions. This exact comparison appears twice in Group C of the book (as questions 3 and 5, word for word); the two have been merged into this single item.

the frequency is unchanged in both cases; the wavelength decreases going into the water and increases coming out of it

29 why a denser liquid needs a smaller disc

$\sin \theta_c = 1 / n \quad \rightarrow \quad n \text{ larger} \quad \rightarrow \quad \theta_c \text{ smaller}$

$r = h \tan \theta_c \quad \rightarrow \quad \theta_c \text{ smaller} \quad \rightarrow \quad r \text{ smaller}$

Worth seeing with numbers: for water ($n = 1.33$) $\theta_c = 48.8^\circ$ and $r = 1.14 h$, while for a liquid with $n = 1.6$ $\theta_c = 38.7^\circ$ and $r = 0.80 h$ — the disc is about 30 % smaller in radius and only half the area.

because a larger n gives a smaller critical angle, which narrows the cone of escaping light, so a smaller disc covers it

30 glass with $\theta_c = 42^\circ$ against water with $\theta_c = 49^\circ$

$\sin \theta_c = 1 / n \quad \rightarrow \quad n = 1 / \sin \theta_c$

glass : $n = 1 / \sin 42^\circ = 1 / 0.6691 = 1.49$

water : $n = 1 / \sin 49^\circ = 1 / 0.7547 = 1.33$

$1.49 > 1.33 \quad \rightarrow \quad \text{the glass is the denser one}$

The rule to carry away: **the smaller the critical angle, the denser the medium**. A diamond, with $n = 2.42$, has a critical angle of only 24° , which is why so much light is trapped inside it and why it sparkles as it does.

the glass is optically denser, because $n = 1/\sin\theta_c$ and the smaller critical angle gives the larger refractive index